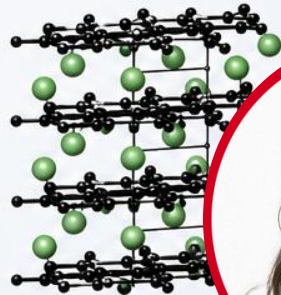
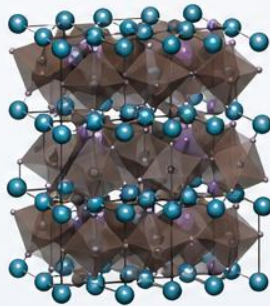
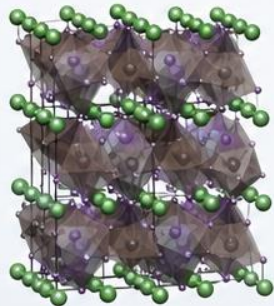
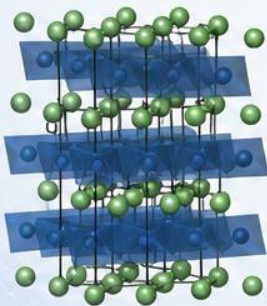


Quantum-Chemical Bonding Analysis of Functional Solid-State Materials (with Emphasis on Batteries)



Dr. Christina Ertural

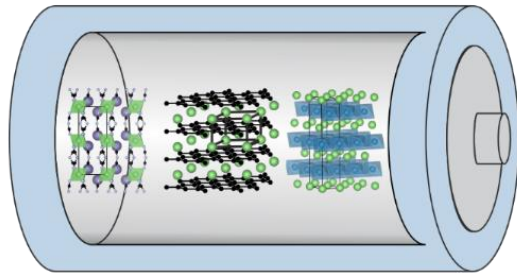
Theoretical Chemistry & Materials Informatics

Outline

- Motivation & short introduction to LOBSTER
- Covalency & Ionicity
 - how it can be used to assess the stability of materials and find new suitable candidates
- Alternative battery materials
 - amorphous materials for anodes
 - transition metal free cathodes

Academic background

- Study of chemistry (B. Sc. & M. Sc. & PhD) at RWTH Aachen University
- Field of research in Prof. Dronskowski's group at Inorganic Institute of RWTH Aachen:
 - *Theoretical and quantum chemistry*
 - > Density functional theory
 - *Chemical bonding* in functional materials
 - > Solid-state chemistry and materials
 - Focus: Method & *C++ code development* concerning chemical bond descriptors in LOBSTER
 - > Mulliken & Löwdin charges, Madelung energy, band-resolved COHP, COBI



Academic background

- Field of research in Prof. George's group at Bundesanstalt für Materialforschung und -prüfung (BAM) Berlin:
 - *Materials informatics*
 - > automation, workflows & machine learning
 - *AutoPLEX*
 - (automatic potential-landscape explorer)
 - > software for generating and benchmarking *machine learning* (ML)-based interatomic potentials
 - Focus: *Python code development* of AutoPLEX from scratch
 - > Materials Project framework

```
from mp_api.client import MPRester
from autoplex.auto.phonons.flows import CompleteDFTvsMLBenchmarkWorkflow

mpr = MPRester(api_key='YOUR_MP_API_KEY')
structure_list = []
benchmark_structure_list = []
mpids = ["mp-22985"]
# you can put as many mpids as needed e.g. mpids = ["mp-22985", "mp-1185319"] for all LiCl ent
mpbenchmark = ["mp-22985"]
for mpid in mpids:
    structure = mpr.get_structure_by_material_id(mpid)
    structure_list.append(structure)
for mpbm in mpbenchmark:
    bm_structure = mpr.get_structure_by_material_id(mpbm)
    benchmark_structure_list.append(bm_structure)

complete_flow = CompleteDFTvsMLBenchmarkWorkflow(
    apply_data_preprocessing=True,
).make(
    structure_list=structure_list, mp_ids=mpids,
    benchmark_structures=benchmark_structure_list, benchmark_mp_ids=mpbenchmark)

complete_flow.name = "tutorial"
autoplex_flow = complete_flow
```



From theory to chemistry...

Theoretical framework

Schrödinger equation

$$\hat{H}\psi = E\psi$$

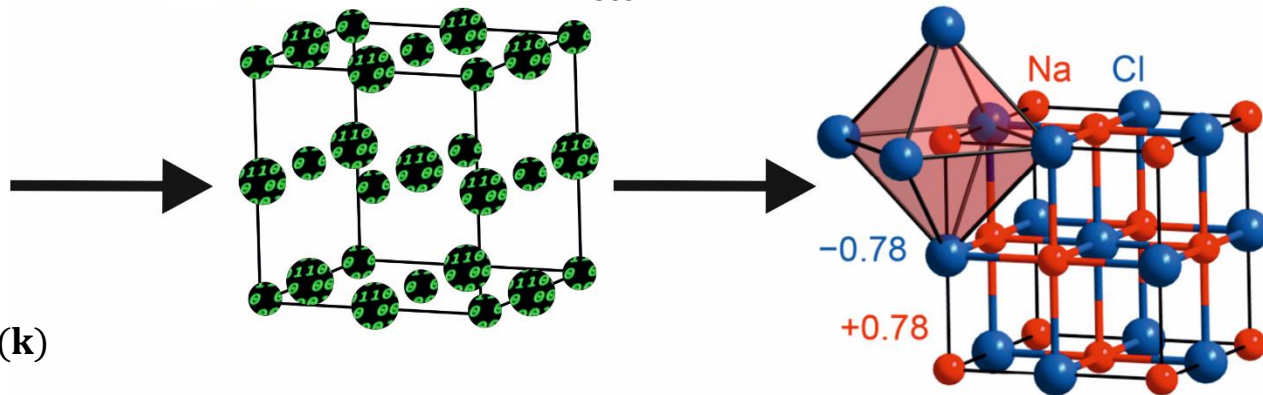
$$|\psi_j(\mathbf{k})\rangle = \sum_{\mu} c_{\mu j}(\mathbf{k}) |\chi_{\mu}(\mathbf{k})\rangle$$

$$\text{COHP}_{\mu\nu} = H_{\mu\nu} \sum_{j,\mathbf{k}} c_{\mu j}^*(\mathbf{k}) c_{\nu j}(\mathbf{k})$$

$$q_A = N - \sum_{\mu \in A} \text{GP}_{\mu}$$



Chemical insights
Structure, charges,
covalency, stability...



Computer simulation
based on experimental
structure models

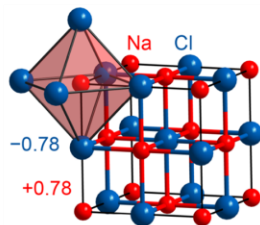
...to (everyday) materials

MULLIKEN and LOEWLIN POPULATION ANALYSIS

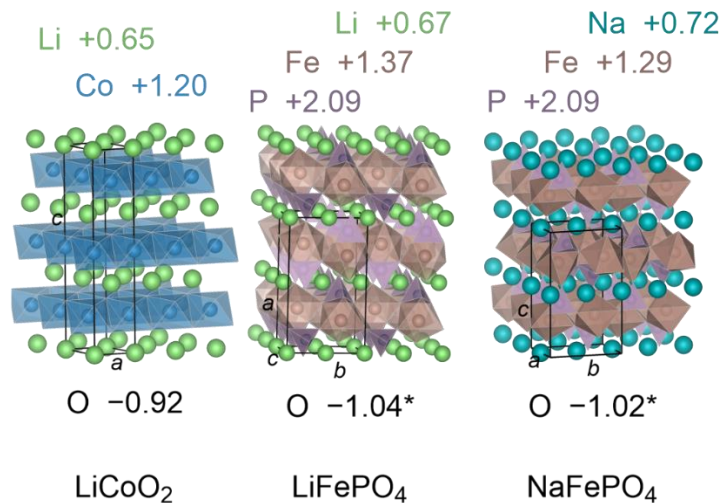
atom#	atom	Mulliken charge	Loewlin charge
1	Na	0.78	0.67
5	Cl	-0.78	-0.67
	total	0.00	0.00

Mulliken and Loewlin gross orbital population

atom#	atom	basisfunction	Mulliken GP	Loewlin GP
1	Na	3s	0.22	0.33
		2p_y	2.00	2.00
		2p_z	2.00	2.00
		2p_x	2.00	2.00
		total	6.22	6.33
5	Cl	3s	1.95	1.87
		3p_y	1.95	1.93
		3p_z	1.95	1.93
		3p_x	1.95	1.93
		total	7.78	7.67

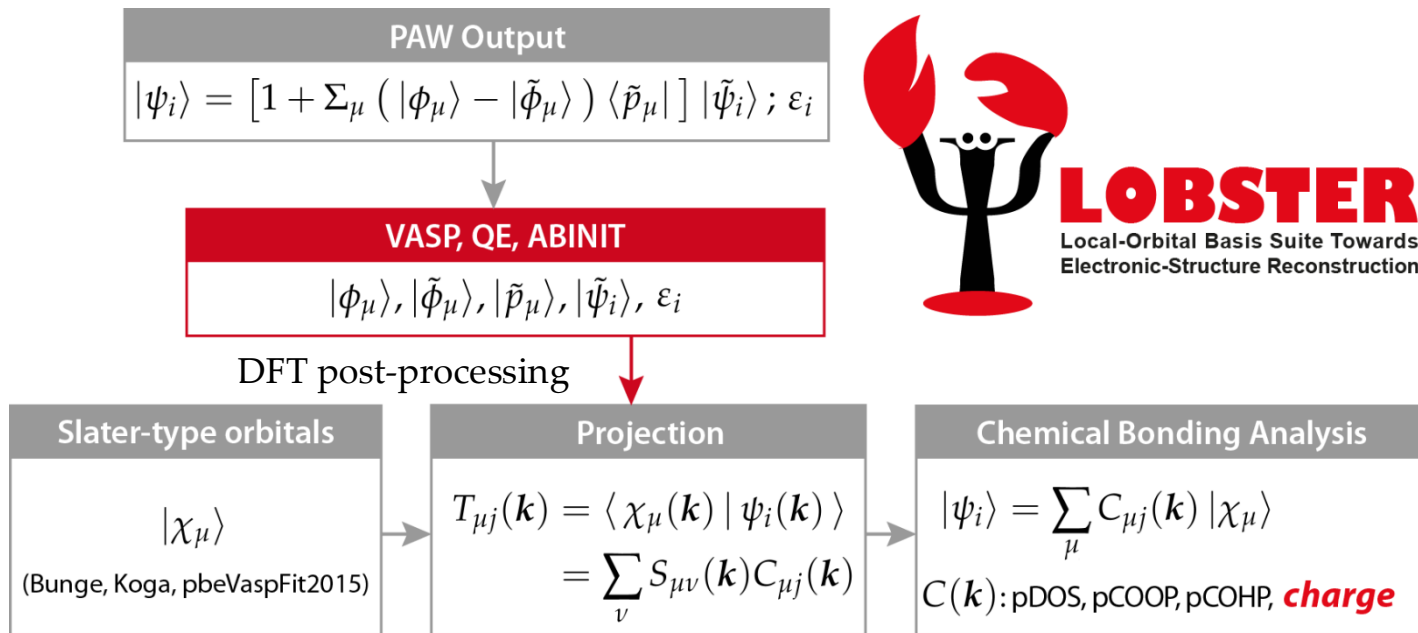


Chemical insights
Structure, charges,
covalency, stability...

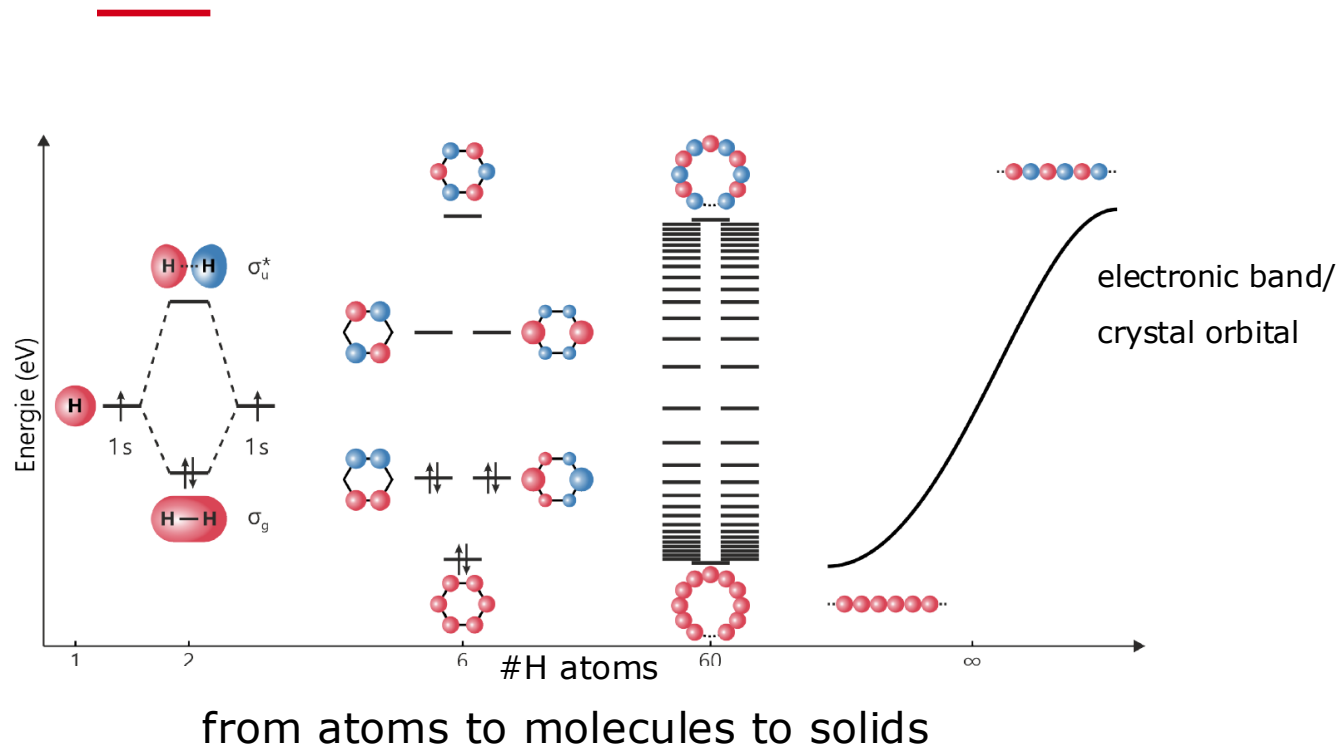


Materials properties
Conductivity, hardness,
(dis)charge behaviour...

The right tool: LOBSTER

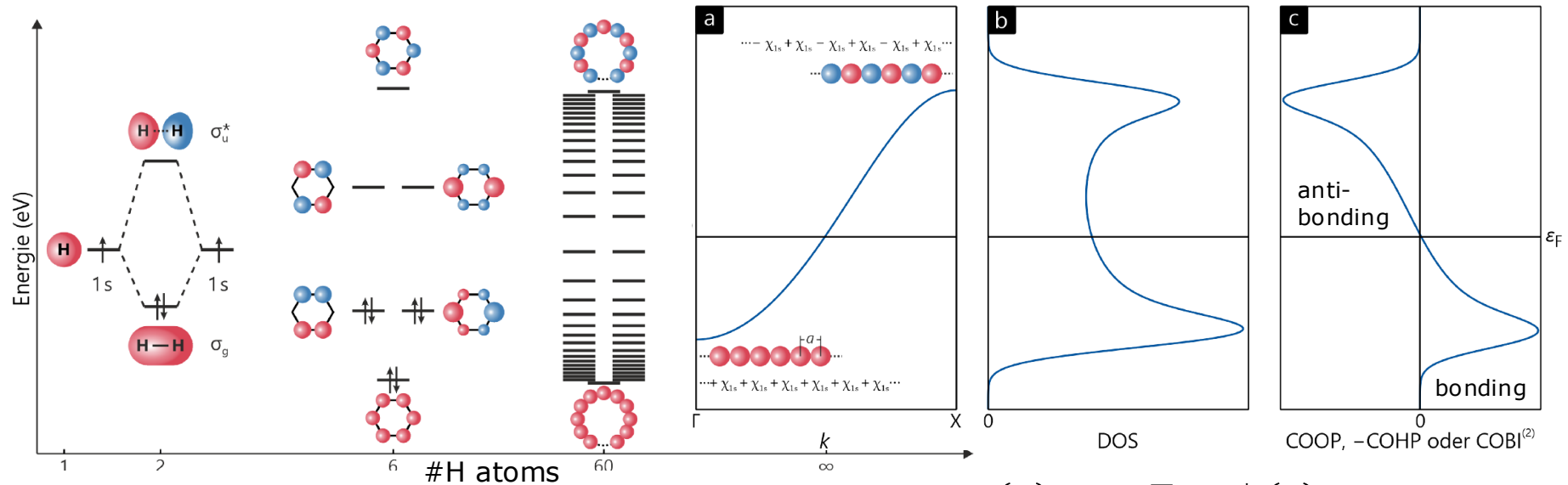


Covalent bond analysis: COOP, COHP



Covalent bond analysis: COOP, COHP

Crystal Orbital Overlap Population & Crystal Orbital Hamilton Population



from atoms to molecules to solids

$$\text{COOP}_{\mu\nu}(E) \sim S_{\mu\nu} \sum_{j,\mathbf{k}} C_{\mu j}^*(\mathbf{k}) C_{\nu j}(\mathbf{k})$$

$$\text{COHP}_{\mu\nu}(E) \sim H_{\mu\nu} \sum_{j,\mathbf{k}} C_{\mu j}^*(\mathbf{k}) C_{\nu j}(\mathbf{k})$$

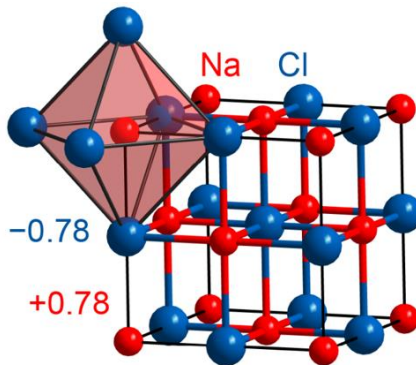
Ionic bond analysis

Mulliken & Löwdin population analysis

- wave function based
- basis set dependent (molecular case)

Popular alternative: Bader charge analysis

- electron density based



LCAO ansatz ($\mathbf{k}$ -space):

$$|\psi_j(\mathbf{k})\rangle = \sum_{\mu} c_{\mu j}(\mathbf{k}) |\chi_{\mu}(\mathbf{k})\rangle$$

Gross (orbital) population:

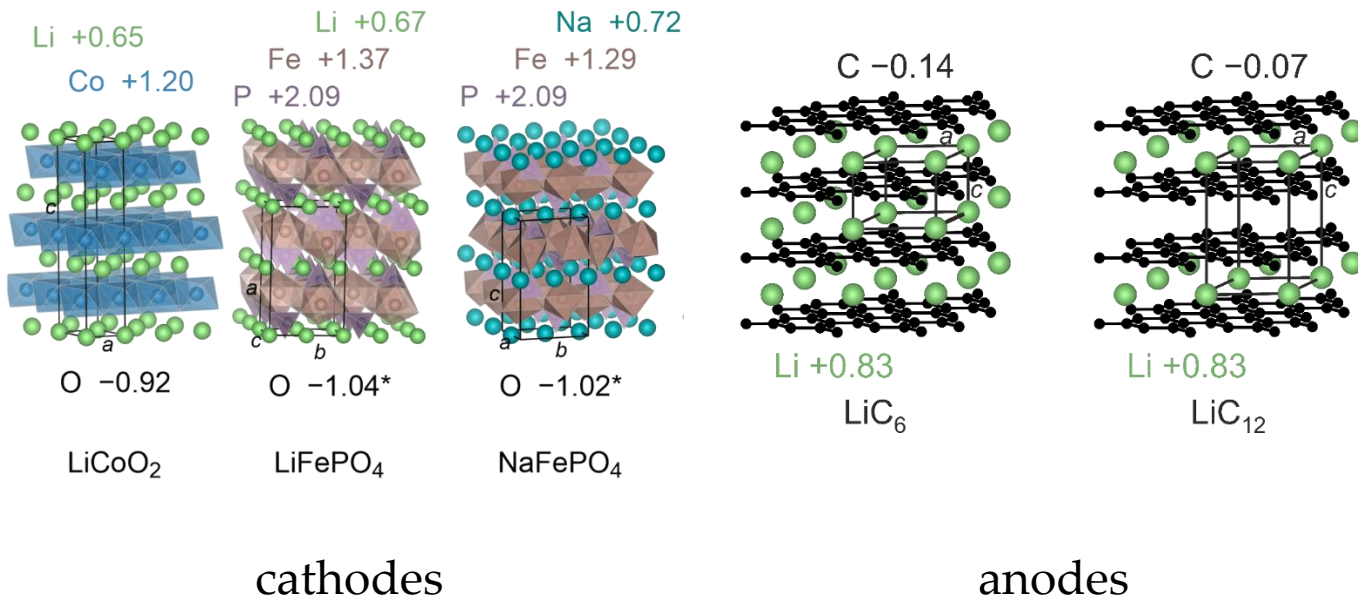
$$GP_{\mu} = \sum_{\mathbf{k}} \sum_{\nu} P_{\mu\nu}(\mathbf{k}) S_{\mu\nu}(\mathbf{k}) w(\mathbf{k})$$

Mulliken/Löwdin charge:

$$q_A = N - \sum_{\mu \in A} GP_{\mu}$$

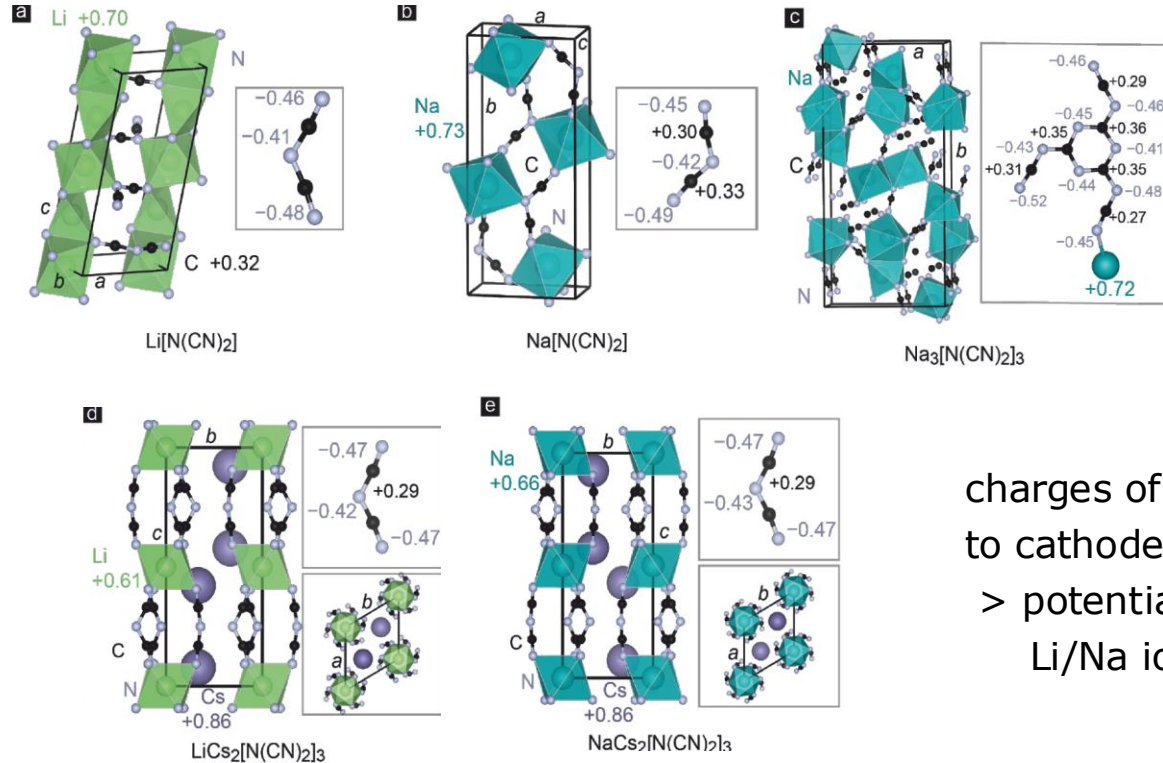
Mulliken & Löwdin charges

- commercially used materials



Ertural, C.; Stoffel, R. P.; Müller, P. C.; Vogt, C. A.; Dronskowski, R., First-Principles Plane-Wave-Based Exploration of Cathode and Anode Materials for Li and Na-ion Batteries involving Complex Nitrogen-based Anions. *Chem. Mater.* **2022**, *34*, 652–668.

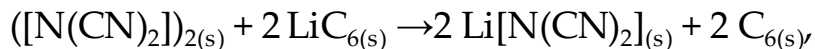
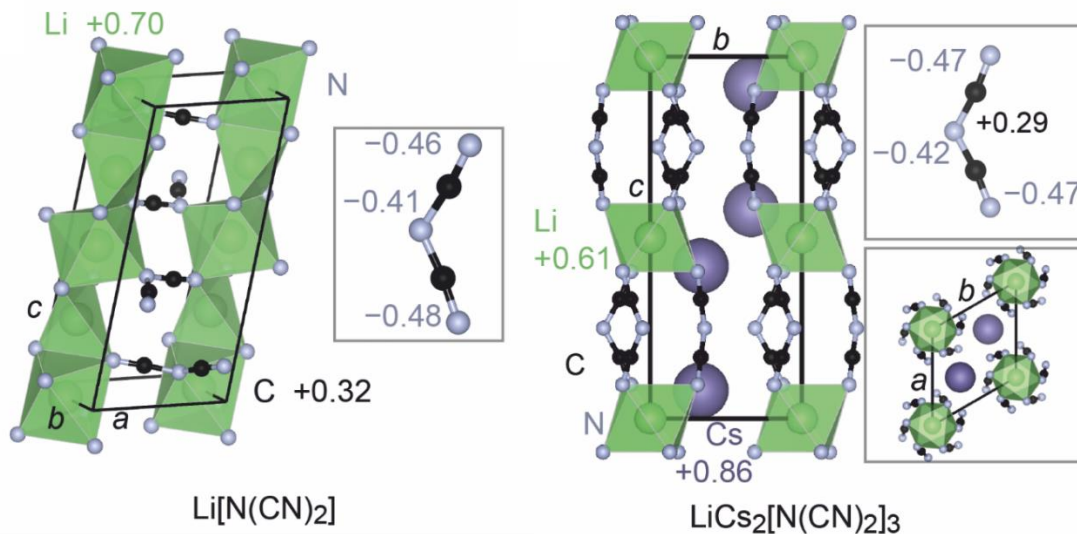
Mulliken & Löwdin charges: Li/Na dicyanamide salts



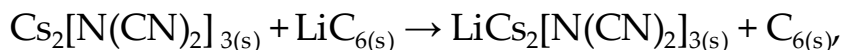
charges of Li and Na comparable
to cathode materials
> potential candidates for
Li/Na ion battery cathodes

Mulliken & Löwdin charges

- new transition metal free candidates



$$\Delta H_r = -390.8 \text{ kJ mol}^{-1} \triangleq \textbf{+4.1 V}$$

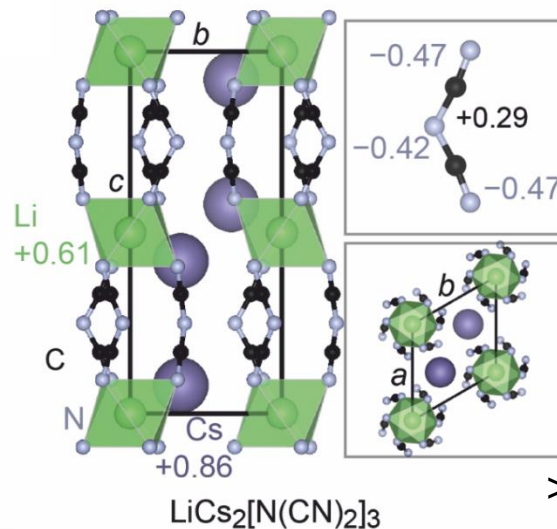
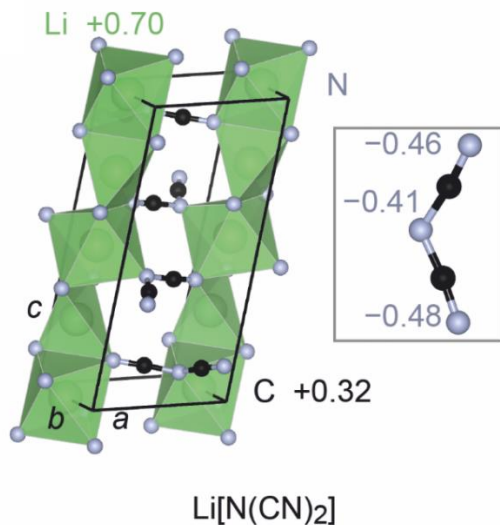


$$\Delta H_r = -444.4 \text{ kJ mol}^{-1} \triangleq \textbf{+4.6 V}$$

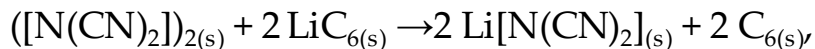
Ertural, C.; Stoffel, R. P.; Müller, P. C.; Vogt, C. A.; Dronskowski, R., First-Principles Plane-Wave-Based Exploration of Cathode and Anode Materials for Li and Na-ion Batteries involving Complex Nitrogen-based Anions. *Chem. Mater.* **2022**, *34*, 652–668.

Mulliken & Löwdin charges

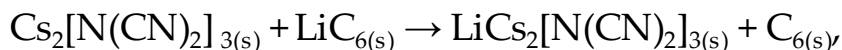
- new transition metal free candidates



> experimental validation ongoing



$$\Delta H_r = -390.8 \text{ kJ mol}^{-1} \triangleq \textbf{+4.1 V}$$

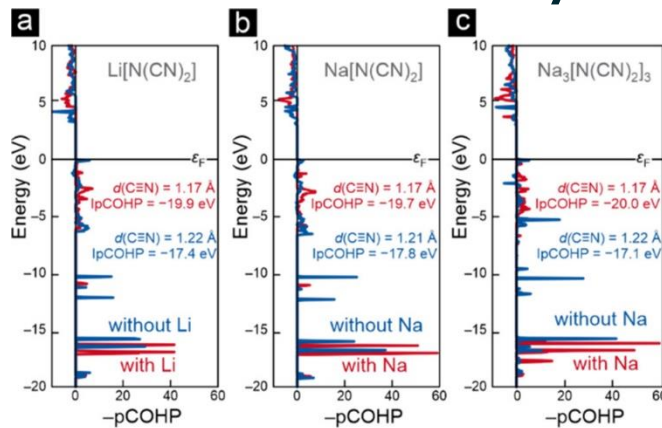


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Ertural, C.; Stoffel, R. P.; Müller, P. C.; Vogt, C. A.; Dronskowski, R., First-Principles Plane-Wave-Based Exploration of Cathode and Anode Materials for Li and Na-ion Batteries involving Complex Nitrogen-based Anions. *Chem. Mater.* **2022**, 34, 652–668.

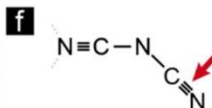
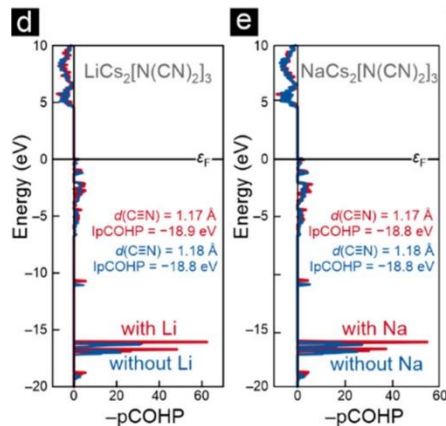
COHP: Deintercalation of Li/Na dicyanamide salts

a lot of
change >



COHP (as measure of chemical bond covalency) plot gives hint on structural deformation and change of chemical environment during deintercalation

almost no
change >



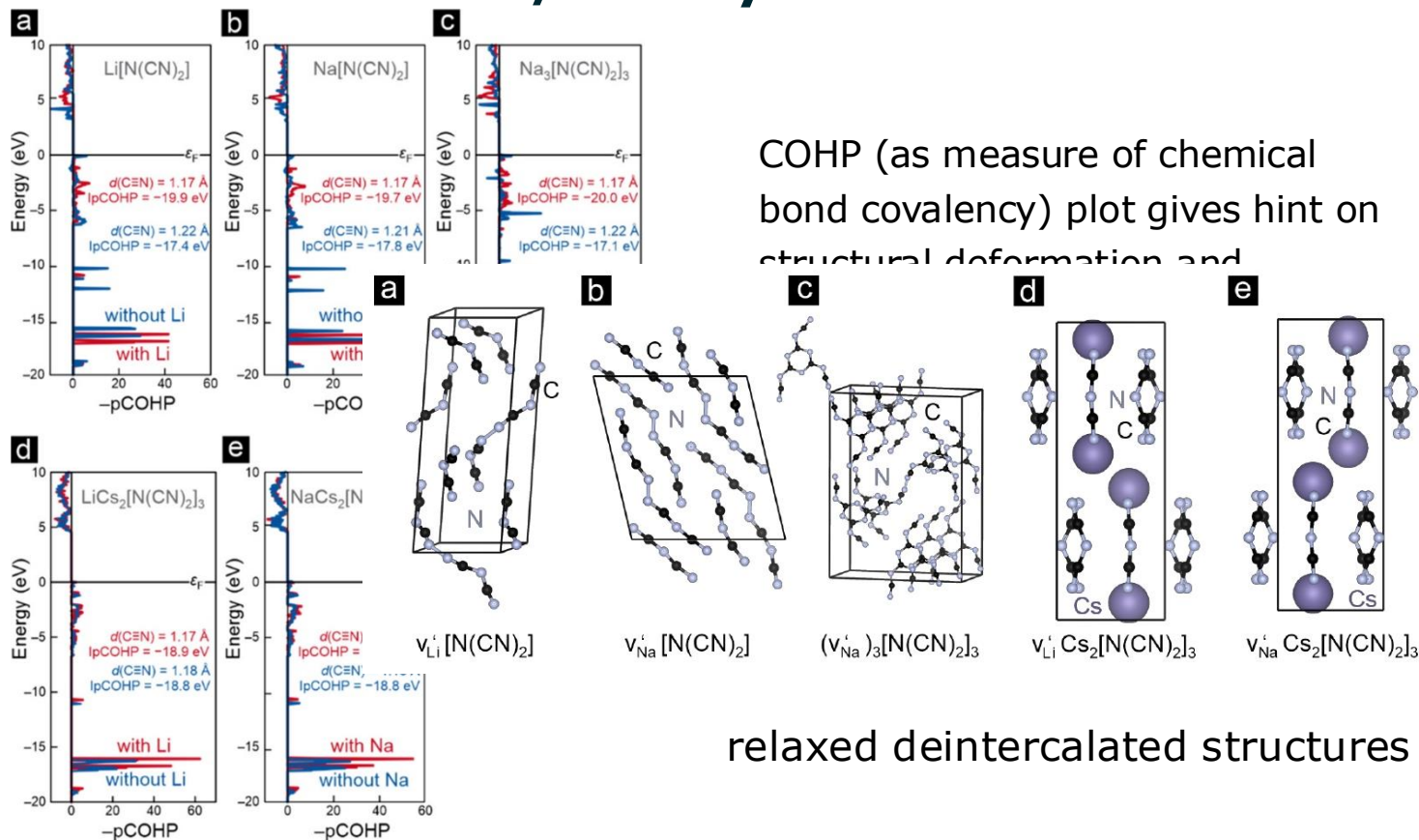
> using COHP to assess stability

Ertural, C.; Stoffel, R. P.; Müller, P. C.; Vogt, C. A.; Dronskowski, R., First-Principles Plane-Wave-Based Exploration of Cathode and Anode Materials for Li and Na-ion Batteries involving Complex Nitrogen-based Anions. *Chem. Mater.* **2022**, 34, 652–668.

COHP: Deintercalation of Li/Na dicyanamide salts

a lot of
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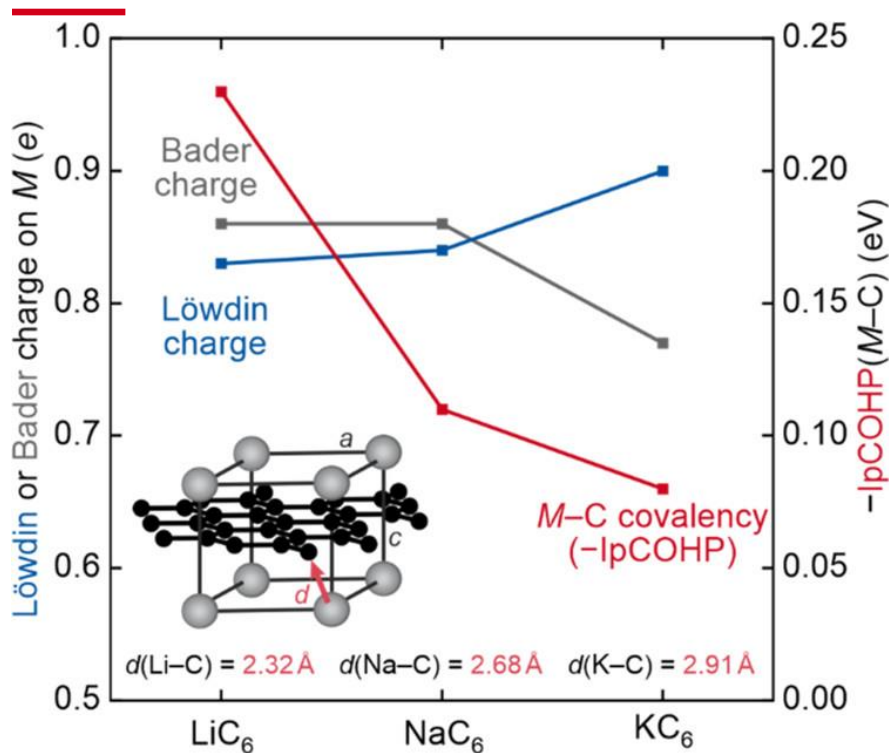
almost no
change >



Ertural, C.; Stoffel, R. P.; Müller, P. C.; Vogt, C. A.; Dronskowski, R., First-Principles Plane-Wave-Based Exploration of Cathode and Anode Materials for Li and Na-ion Batteries involving Complex Nitrogen-based Anions. *Chem. Mater.* **2022**, 34, 652–668.

Charges combined with COHP

- The intercalation problem of Na in (pristine) graphite



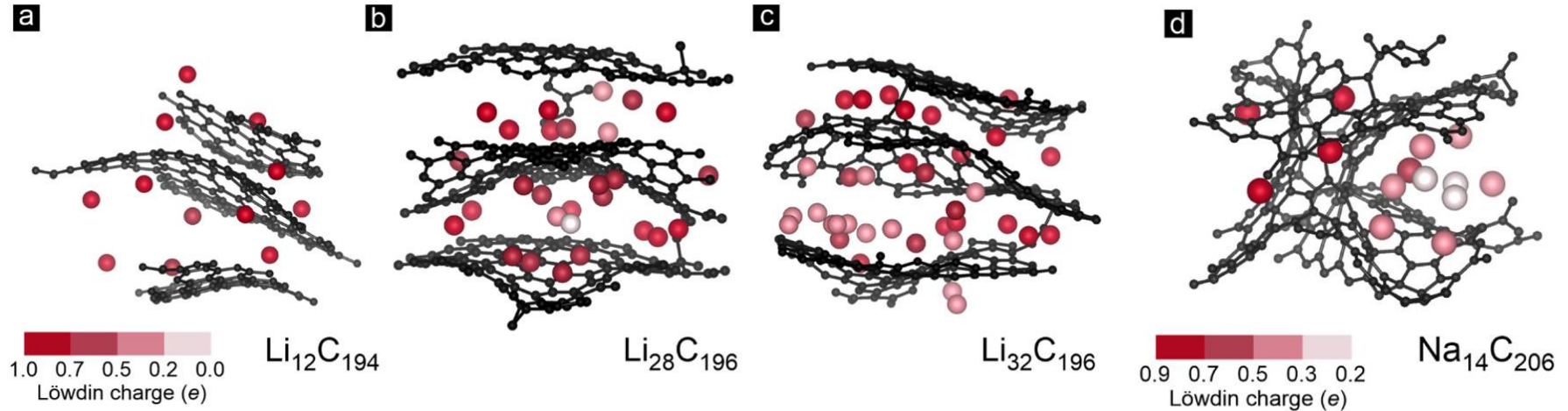
- > only Li is known to intercalate into pristine graphite, yielding LiC₆
- > Na does not intercalate at all
- > K favors the composition KC₈ (but stability issues known)
- > **why?**

Literature: LiC₆ is stabilized by an additional covalent contribution in M-C binding energy

- > confirmed by present study

Mulliken & Löwdin charges

- Amorphous carbonaceous materials as alternative to Li/graphite anodes and to solve the intercalation problem of Na in (pristine) graphite

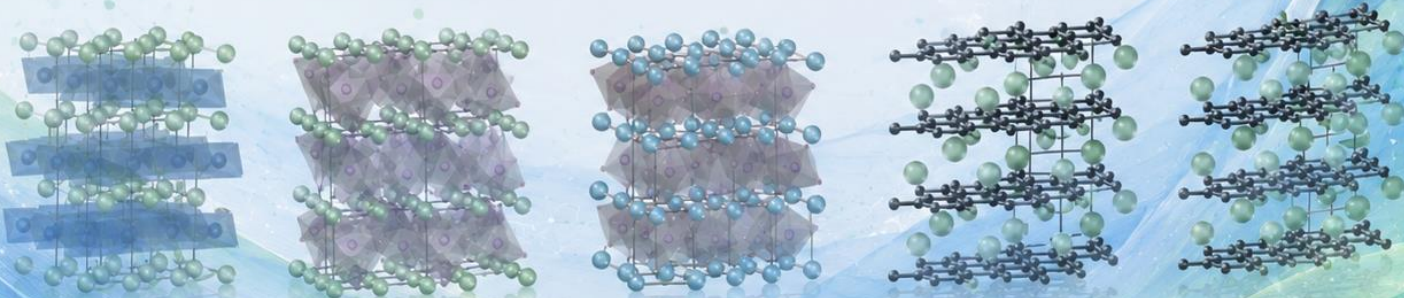


> good Coulombic (charging/discharging) behaviour
expected estimated from the Löwdin charges

Ertural, C.; Stoffel, R. P.; Müller, P. C.; Vogt, C. A.; Dronskowski, R., First-Principles Plane-Wave-Based Exploration of Cathode and Anode Materials for Li and Na-ion Batteries involving Complex Nitrogen-based Anions. *Chem. Mater.* **2022**, *34*, 652–668.

Summary

- Commercial battery materials were examined using chemical bonding indicators
- Insights were used to look for alternative materials
 - transition metal free cathode materials
 - graphite-like amorphous anode materials to intercalate Na



Summary

- Commercial battery materials were examined using chemical bonding indicators
- Insights were used to look for alternative materials
 - transition metal free cathode materials
 - graphite-like amorphous anode materials to intercalate Na

Thank you for your kind attention!

Check out quantumchemist.de (or scan the QR code) to learn more about my background!

